Camunda DMN Decision Table HIT policy UNIQUE

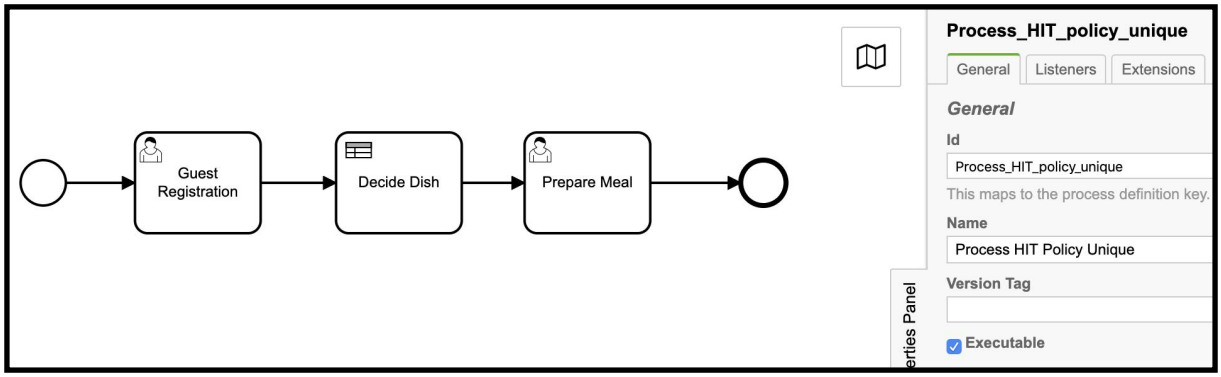
In this session we will try to understand what is decision table with HIT policy UNIQUE

Each row in a decision table is a rule. Hit policy "Unique" will typically make it easy to build a decision table which makes sure that your rules are "complete" - in the sense that the rules do not overlap but cover all possible input values - so that you do not "forget" anything.

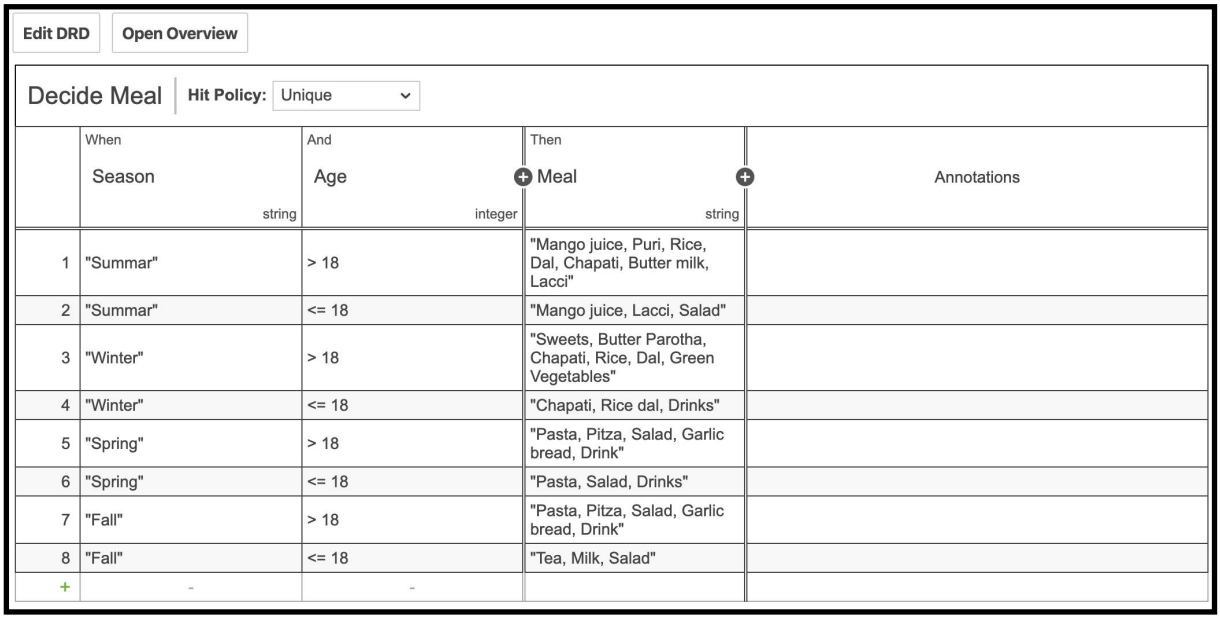
It is an advantage that Unique hit policy makes your decision logic invalid in case you violate its requirement that your table rules never "overlap": after all you must not produce ambiguous results.

To demonstrate the user of HIT policy UNIQUE, We will model below process and decision table.

1. Process definition (BPMN)



2. Decision table (DMN)

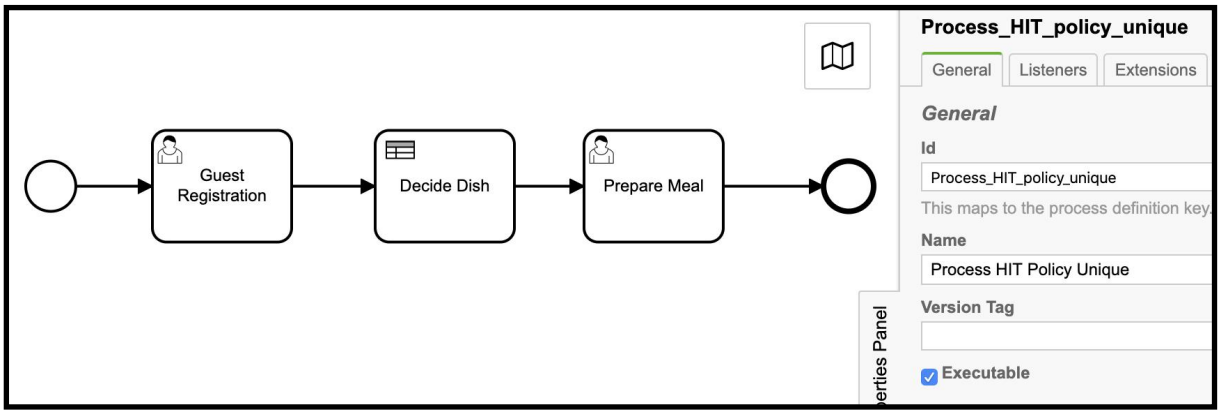


Edit DRD | Open Overview

Decide Meal | Hit Policy: Unique

	When Season string	And Age integer	Then Meal string	Annotations
1	"Summar"	> 18	"Mango juice, Puri, Rice, Dal, Chapati, Butter milk, Lacci"	
2	"Summar"	<= 18	"Mango juice, Lacci, Salad"	
3	"Winter"	> 18	"Sweets, Butter Parotha, Chapati, Rice, Dal, Green Vegetables"	
4	"Winter"	<= 18	"Chapati, Rice dal, Drinks"	
5	"Spring"	> 18	"Pasta, Pitza, Salad, Garlic bread, Drink"	
6	"Spring"	<= 18	"Pasta, Salad, Drinks"	
7	"Fall"	> 18	"Pasta, Pitza, Salad, Garlic bread, Drink"	
8	"Fall"	<= 18	"Tea, Milk, Salad"	
+	-	-		

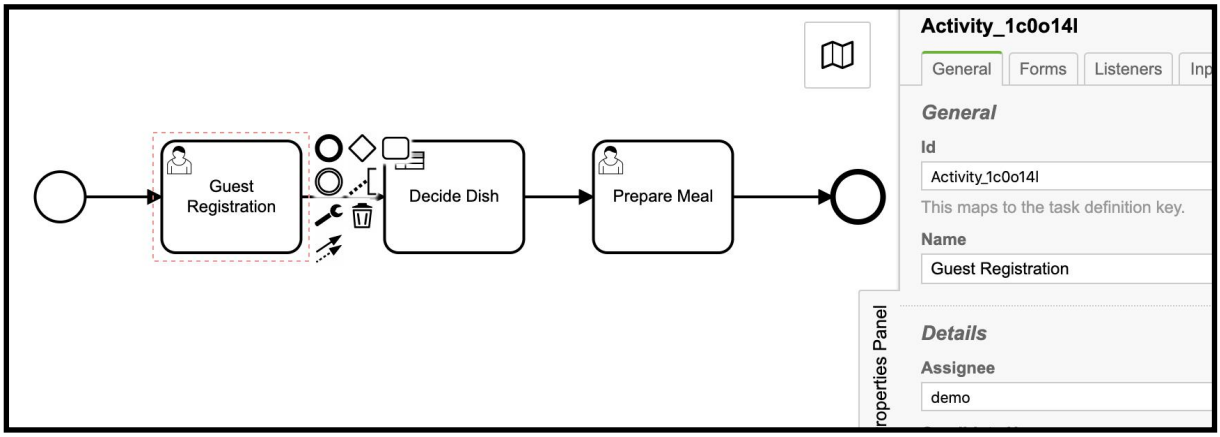
3. Process definition properties



Id: Process_HIT_policy_unique

Name: Process HIT Policy Unique

3.1. User task: Guest Registration

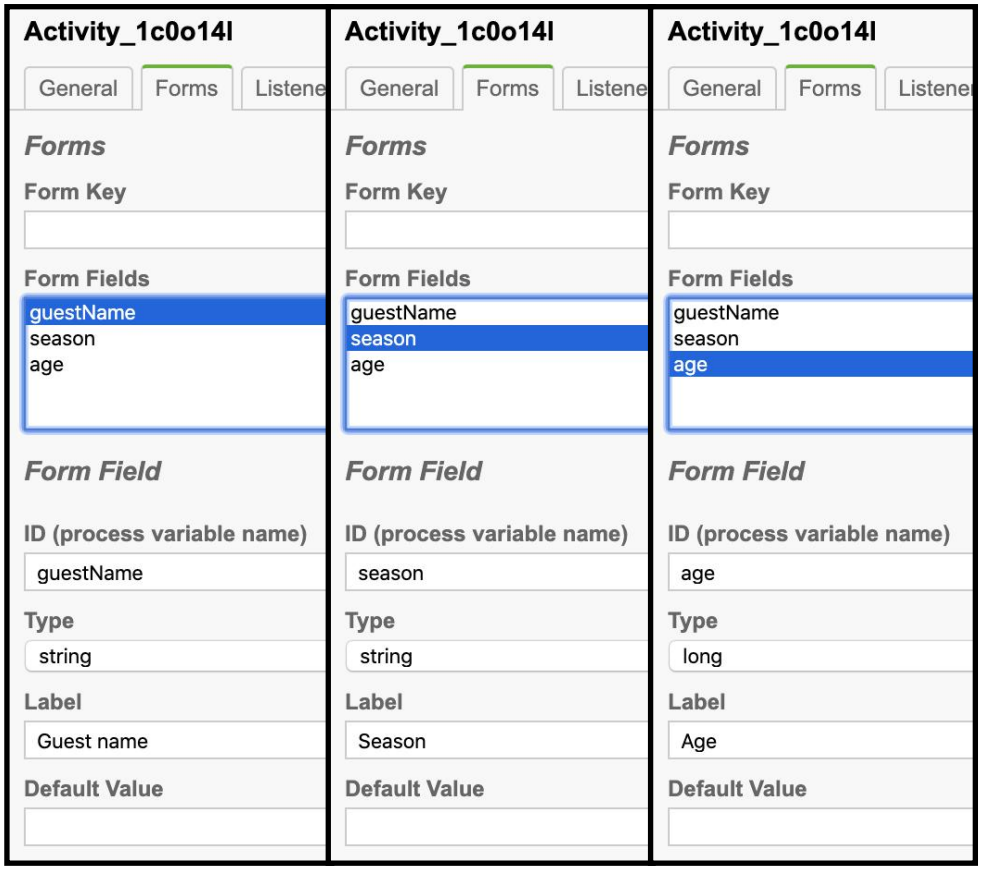


Id: Activity_1c0o14l

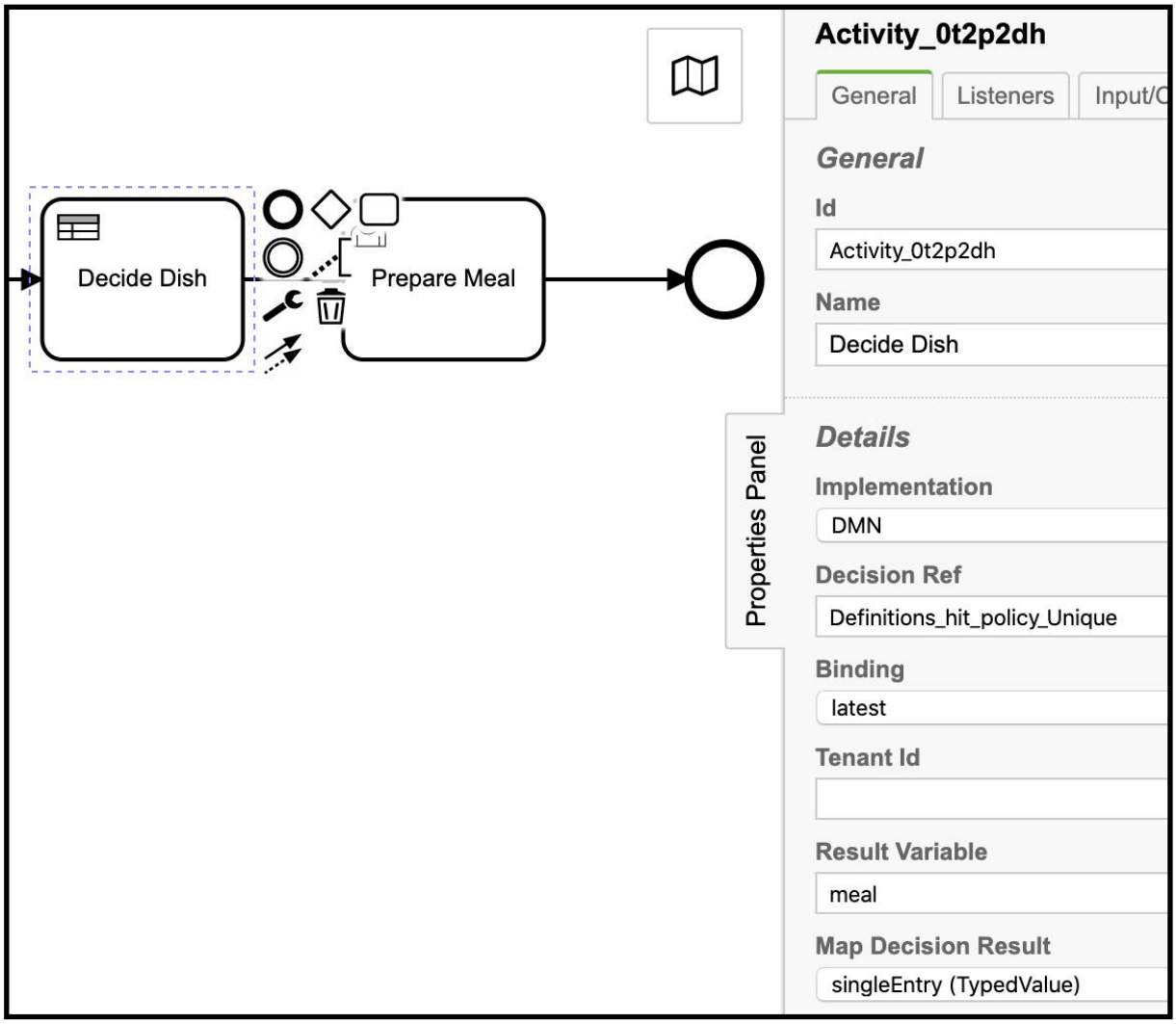
Name: Guest Registration

Assignee: demo

Task form



3.2. Business rule task: Decide Dish



Id: Activity_1c0o14l

Name: Guest Registration

Implementation: DMN

Decision Ref: Definitions_hit_policy_Unique

Binding: latest

Result Variable: meal

Map Decision Result: singleEntry (TypedValue)

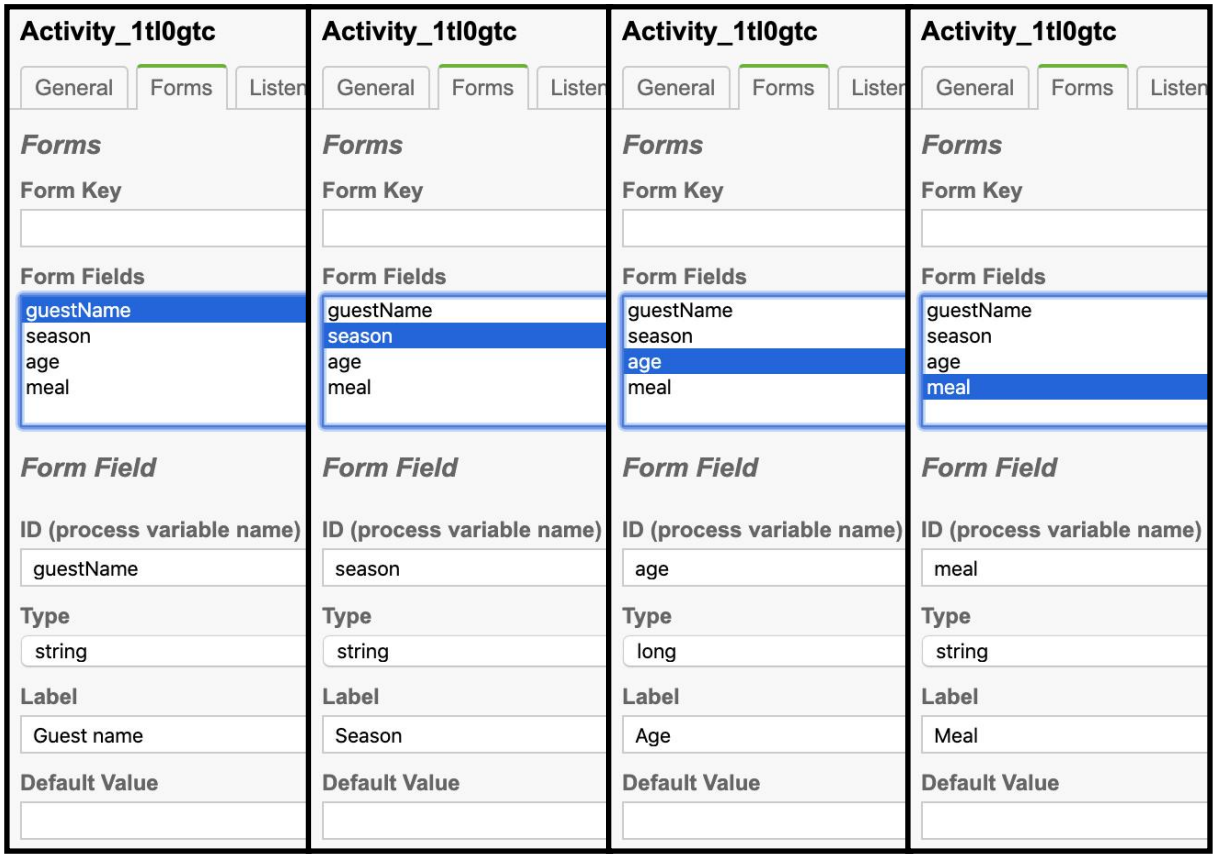
3.3. User task: Prepare Meal

Id: Activity_1tl0gtc

Name: Prepare Meal

Assignee: demo

Task form



4. Decision table:

Edit DRD | Open Overview

Decide Meal | Hit Policy: Unique

	When Season string	And Age integer	Then Meal string	Annotations
1	"Summar"	> 18	"Mango juice, Puri, Rice, Dal, Chapati, Butter milk, Lacci"	
2	"Summar"	<= 18	"Mango juice, Lacci, Salad"	
3	"Winter"	> 18	"Sweets, Butter Parotha, Chapati, Rice, Dal, Green Vegetables"	
4	"Winter"	<= 18	"Chapati, Rice dal, Drinks"	
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8	"Fall"	<= 18	"Tea, Milk, Salad"	
+	-	-		

Properties:

Definitions_hit_policy_Unique

General

General

Id

Definitions_hit_policy_Unique

This maps to the decision definition key.

Name

Decide Meal

Version Tag

History Configuration

History Time To Live

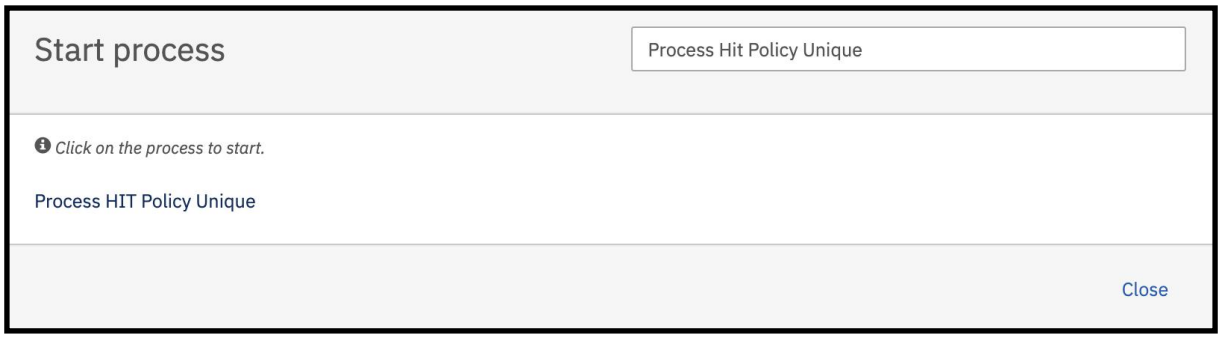
Id: Definitions_hit_policy_Unique

Name: Decide Meal

5. Deploy process definition and decision table.

6. Testing the process definition and DMN

Start the process definition



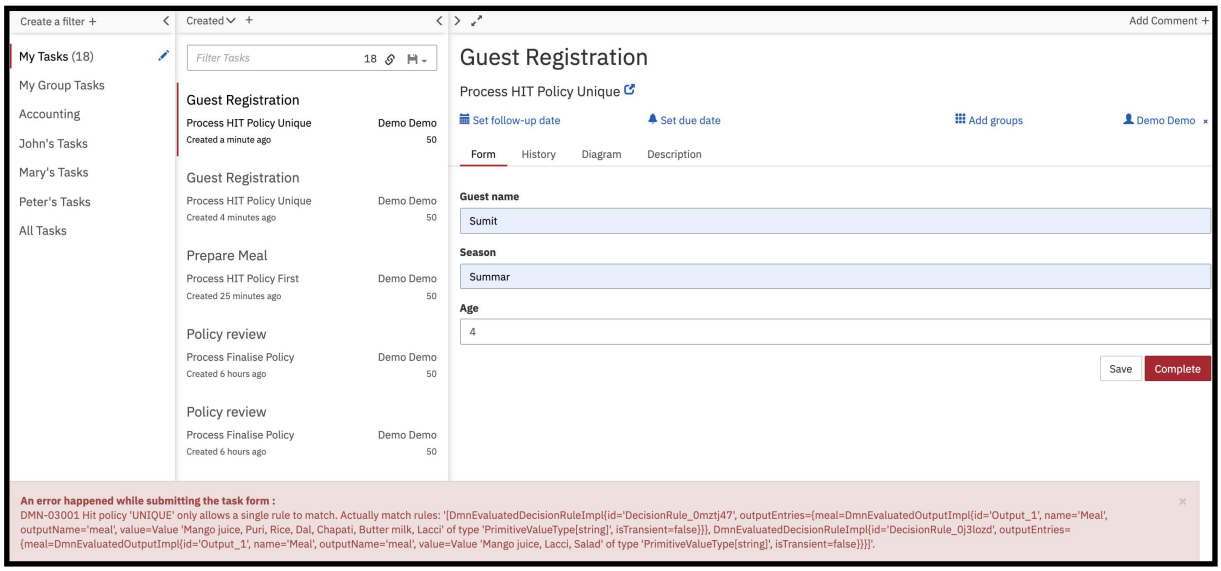
Hit **Process HIR Policy Unique** and press Start button to start process.

As per process definition, user task Guest Registration will get created.

Fill-up the task form and press Complete button and then as per process definition business rule task get executed and will produce single output based on provided data by task and same output result can be viewed on newly created user task Prepare Meal.

It can be seen that business rule task has invoked decision table and produces output Meal = Mango juice, Lacci, Salad

Note: In decision table if you have two rules satisfied with provided input data then error will be thrown as below



Tomcat console error logs

14-Sep-2020 18:31:05.184 WARNING [http-nio-8080-exec-1]
org.camunda.commons.logging.BaseLogger.logWarn ENGINE-REST-HTTP500
org.camunda.bpm.dmn.engine.impl.hitpolicy.DmnHitPolicyException: DMN-03001 Hit policy
'UNIQUE' only allows a single rule to match. Actually match rules:
'[DmnEvaluatedDecisionRuleImpl{id='DecisionRule_0mztj47',
outputEntries={meal=DmnEvaluatedOutputImpl{id='Output_1', name='Meal', outputName='meal',
value=Value 'Mango juice, Puri, Rice, Dal, Chapati, Butter milk, Lacci' of type
'PrimitiveValueType[string]', isTransient=false}}},
DmnEvaluatedDecisionRuleImpl{id='DecisionRule_0j3lozd',
outputEntries={meal=DmnEvaluatedOutputImpl{id='Output_1', name='Meal', outputName='meal',
value=Value 'Mango juice, Lacci, Salad' of type 'PrimitiveValueType[string]', isTransient=false}}}]'.